

3.5 Converse of the Alternate Interior Angles Theorem

Lesson Introduction

 Benchmarks	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles.			 Learning Targets	- I can write and use the converse of the postulate or theorem. - I can use postulates and theorems about parallel lines to determine if lines are parallel.
	MA.912.LT.4.3 Identify and accurately interpret “if...then,” “if and only if,” “all,” and “not” statements. Find the converse, inverse, and contrapositive of a statement.				
 Benchmark Coherence	Building On MA.8.GR.1.4: Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.			Working Toward N/A	
 Essential Questions	- How do you write the converse of a statement? - What angle relationships can be used to determine if two lines are parallel? - What angle relationships cannot be used to determine if two lines are parallel?				
 Lesson Narrative	The lesson connects the postulates and theorems learned in the previous lessons to their converses. Students will determine and justify whether lines are parallel given the relationship of angle pairs formed by two lines intersected by a transversal. Students will explore which information is sufficient to prove lines are parallel and which is not.				
 Component Summary	3.5.1 Warm-Up (~5 min.)	Notice and Wonder about lines in a video game (<i>SE p. 150</i>)	3.5.3 Activity (15-20 min.)	Gallery Walk exploring the converses of parallel line postulates and theorems (<i>SE p. 151</i>)	
	3.5.2 Exploration (5-8 min.)	Conjecture if given angle relationships are sufficient to determine lines are parallel (<i>SE p. 150</i>)	3.5.4 Guided Practice (10-12 min.)	Determine if lines are parallel using angle relationships (<i>SE p. 152</i>)	
	3.5.5 Wrap-Up (~5 min.)	Students summarize their learning (<i>SE p. 153</i>)			
 Resources	Glossary		Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Congruent - Converse - Parallel Lines		(Optional) Chart Paper for Gallery Walk	Consider having a Word Wall or Anchor Chart in the classroom that includes images of the various angle relationships created by parallel lines intersected by a transversal.	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may incorrectly say lines are parallel based on linear pairs being supplementary and vertical angles being congruent.	When students struggle justifying lines are parallel when given multiple transversals, consider having students make a list of the angles formed by each transversal prior to answering any questions. This move will help students see the angle pairs before determining the relationships.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
 Differentiate Instruction	Take Turns In 3.5.4 Guided Practice, have students employ the Take Turns routine when sharing their response in the table. Using this routine limits the likelihood that one student takes over while the other student does not participate. If students disagree with their partner’s answer, they should provide a rationale for their disagreement and resolve it through discussion.	MA.K12.MTR.4.1: Discussion and Reflection Throughout the lesson, students are asked to communicate mathematical ideas, vocabulary involving angle relationships, and their methods for justifying if two lines are parallel. During the Gallery Walk in the 3.5.3 Activity, students will also analyze the statements of others to determine if it is correct and/or what corrections need to be made.
	The 3.5.3 Activity provides both auditory and visual entry points. Students discuss the converse of each postulate and theorem with a partner and provide a visual example.	
Lesson Notes:		

3.5.1 Warm-Up: Notice and Wonder

Component Overview: Students consider a video game screen and write down what they notice and wonder.



3.5.1 Warm-Up: Notice and Wonder

The game *Donkey Kong* was released for sale in the U.S. in 1981 by Nintendo, a Japanese playing card and toy company turned video game developer. Mario, circled in yellow, is shown in the image running and climbing up the platforms and ladders.



- What do you notice about the different platforms Mario is running on?
Answers will vary. They represent line segments that are not parallel to each other.
- What do you wonder?
Answers will vary. Can Mario climb any of the ladders?
- How can you determine if any of the platforms or ladders are parallel?
Answers will vary. I can determine if the ladders are parallel by measuring the angles formed by the platforms and the ladders to determine the angle measure relationships.



Consider having students compare what they noticed in question 1 with their partner and write down one thing that their partner noticed that they did not.

If time allows, consider asking students to share their responses to question 3 with a partner or with the whole group as a transition to the next component.

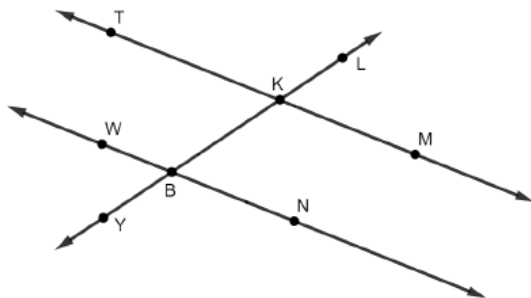
3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

Component Overview: Students will work with a partner to conjecture if there is enough information to determine if two lines are parallel from various pieces of given information.



3.5.2 Exploration: Congruent Angles Formed by Two Lines and a Transversal

1. $\overleftrightarrow{TM}$, $\overleftrightarrow{WN}$, and $\overleftrightarrow{YL}$ are shown. $\overleftrightarrow{YL}$ intersects $\overleftrightarrow{TM}$ and $\overleftrightarrow{WN}$.



Work with your partner to conjecture if the given information in each row of the table is sufficient to show that the lines are parallel.

Angles	Type of Angles	Sufficient?
$\angle TKL \cong \angle WBK$	Corresponding angles	☐ Yes ☐ No
$\angle MKB \cong \angle WBK$	Alternate interior angles	☐ Yes ☐ No
$\angle MKB \cong \angle TKL$	Vertical angles	☐ Yes ☐ No
$\angle TKL \cong \angle YBN$	Alternate exterior angles	☐ Yes ☐ No
$\angle WBK$ and $\angle NBK$ are supplementary.	Linear pair	☐ Yes ☐ No
$\angle TKL$ and $\angle WBY$ are supplementary.	Consecutive exterior angles	☐ Yes ☐ No



For the angles that do not have sufficient evidence to show that the lines are parallel, have students explain their reasoning. Listen for evidence that includes the angles being formed by one line and the transversal, and not two lines and a transversal.



At the conclusion of 3.5.2 Exploration, consider guiding students through a discussion of their answers, if time allows. This can make for a rich mathematical discussion, as well as address any potential misconceptions.

3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

Component Overview: In this component, students write converses of postulates or theorems and draw pictorial representations.



3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems

1. Work with your partner or group to write the converse of the postulate or theorem your group is assigned. Then, draw a picture to represent the relationship described.

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate interior angles are congruent.	<i>If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.</i>
Picture	



Consider structuring 3.5.3 Activity as a Gallery Walk. Assign each group or student pair a postulate or theorem and have them write the converse and draw a pictorial representation on large sheets of paper. Once done, have the students participate in a Gallery Walk where each group presents their representation. Students record the representations in their workbooks.

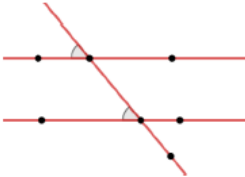


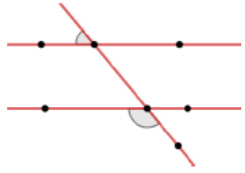
If students do not recall how to write a converse statement, consider prompting them to revisit Lesson 4 from Unit 2 in their workbook.



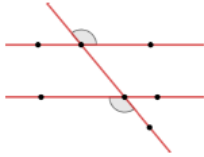
When two lines are intersected by a transversal, if the angle pair relationships of parallel lines and a transversal occur, then the lines are parallel. Each angle pair relationship formed by parallel lines intersected by a transversal postulate and theorem can be written as converses.

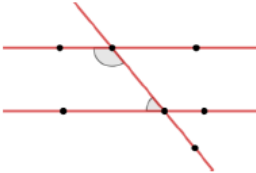
3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems (continued)

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then corresponding angles are congruent.	<i>If two lines are intersected by a transversal so that corresponding angles are congruent, then the lines are parallel.</i>
Picture	
	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive exterior angles are supplementary.	<i>If two lines are intersected by a transversal so that consecutive exterior angles are supplementary, then the lines are parallel.</i>
Picture	
	

3.5.3 Activity: Converses of Parallel Lines Postulates and Theorems (continued)

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then alternate exterior angles are congruent.	<i>If two line are intersected by a transversal so that alternate exterior angles are congruent, then the lines are parallel.</i>
Picture	
	

Postulate or Theorem	Converse
If two parallel lines are intersected by a transversal, then consecutive interior angles are supplementary.	<i>If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.</i>
Picture	
	

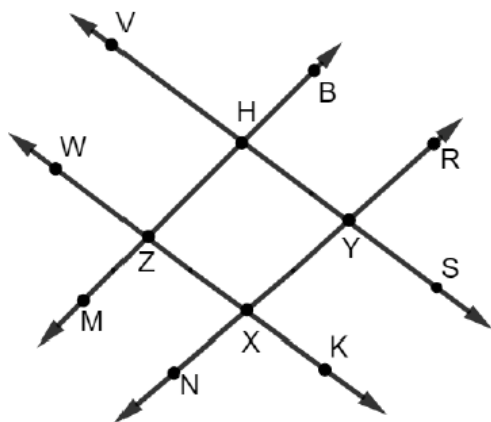
3.5.4 Guided Practice: Determining if Lines Are Parallel

Component Overview: Students use the converse of the angle pair relationship theorems and postulates to determine which lines are parallel, including providing justification.



3.5.4 Guided Practice: Determining if Lines are Parallel

1. $\overleftrightarrow{VS}$, $\overleftrightarrow{WK}$, $\overleftrightarrow{MB}$, and $\overleftrightarrow{NR}$ are shown where $m\angle VHB = 98^\circ$, $m\angle RYS = 78^\circ$, $m\angle NXK = 102^\circ$, and $m\angle WZM = 82^\circ$.



Which pair(s) of lines are parallel? Justify your answer.

Answers will vary. $\overleftrightarrow{VS}$ and $\overleftrightarrow{WK}$ are parallel because $\angle VHB$ and $\angle WZM$ are supplementary consecutive exterior angles formed by the two lines and transversal $\overleftrightarrow{MB}$.



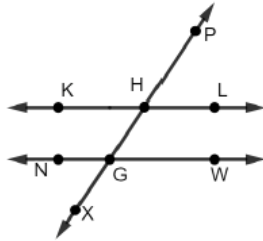
Use observational data from 3.5.2 Exploration and 3.5.3 Activity to determine how much guidance is needed to bring together key ideas in 3.5.4 Guided Practice. If necessary, consider modeling question 1 to address common misconceptions shown in prior components.



For students who finish quickly, consider having them explain why lines $\overleftrightarrow{BM}$ and $\overleftrightarrow{RN}$ are not parallel? Students should provide at least one angle relationship that supports their answer.

3.5.4 Guided Practice: Determining if Lines Are Parallel (continued)

2. $\overleftrightarrow{KL}$, $\overleftrightarrow{NW}$, and $\overleftrightarrow{PX}$ are shown.



Determine if the given condition could be used to justify that $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$. Then, justify your answer.

Condition	Is $\overleftrightarrow{KL} \parallel \overleftrightarrow{NW}$?	Justification
$\angle PHL \cong \angle WGX$	☐ Yes ☐ No	<i>Congruent consecutive exterior angles are not a justification for proving parallel lines.</i>
$m\angle KHP + m\angle NGX = 180^\circ$	☐ Yes ☐ No	<i>Consecutive exterior angles are supplementary.</i>
$\angle KHG \cong \angle WGH$	☐ Yes ☐ No	<i>Alternate interior angles are congruent.</i>
$\angle NGX$ and $\angle WGX$ are supplementary.	☐ Yes ☐ No	<i>$\angle NGX$ and $\angle WGX$ are a linear pair, which is not a justification for proving parallel lines.</i>
$\angle LHG \cong \angle KHP$	☐ Yes ☐ No	<i>$\angle LHG$ and $\angle KHP$ are vertical angles, which is not a justification for proving parallel lines.</i>



Take Turns

For question 2, consider having students employ the Take Turns routine when sharing their response in the table. Using this routine limits the likelihood that one student takes over while the other student does not participate. If students disagree with their partner's answer, they should provide a rationale for their disagreement and resolve it with a discussion.

3.5.5 Wrap-Up: Cool Down

Component Overview: Students write a short summary of their learning in the lesson. Student responses can be used to identify misconceptions and plan for upcoming instruction.



3.5.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed today's class summarizing what you learned.

Answers will vary.